Unit: 6 Unit 6: Systems of Linear Equations and Inequalities **Name:** ______________
Topic: Solving Systems by Graphing **Date:** __________

Homework (Habanero)

Q1. A medical trial tests the effects of two drugs, x and y, on blood pressure. The trial's results can be modeled by the system: $2x + 3y = 12$ and $x - 2y = -3$. What is the intersection point of these two lines?
(A) $(0, 4)$
(B) $(3, 2)$
(C) $(2, 1)$
(D) $(1, 3)$
(E) $(4, 0)$

Q2. The growth rate of a certain cell culture is modeled by the equation $y = 2x + 5$. If the initial cell count is 10, what is the y-intercept of this line?
(A) $(0, 5)$
(B) $(0, 10)$
(C) $(0, 15)$
(D) $(5, 0)$
(E) $(10, 0)$

Q3. A pharmacist has 480 tablets of a certain medication. The pharmacist wants to package them in bottles of 12 and 8 tablets. If x represents the number of bottles with 12 tablets and y represents the number of bottles with 8 tablets, which system of equations models this situation?
(A) $x + y = 480$, $12x + 8y = 4800$
(B) $x + y = 40$, $12x + 8y = 480$
(C) $2x + y = 40$, $12x + 8y = 480$
(D) $x - y = 40$, $12x - 8y = 480$
(E) $x + 2y = 40$, $12x + 16y = 480$

Q4. The cost of producing x units of a medication is $2x + 1000$ dollars, and the revenue from selling x units is $5x - 2000$ dollars. If the profit P is the revenue minus the cost, what is the break-even point, where $P = 0$?
(A) $x = 300$
(B) $x = 400$
(C) $x = 600$
(D) $x = 800$
(E) $x = 1000$

Q5. The graph of the system $y = x + 3$ and $y = -2x - 2$ has an intersection point at:
(A) $(0, 3)$
(B) $(1, 4)$
(C) $(-1, 2)$
(D) $(-2, 1)$
(E) $(-5, -7)$

Q6. What is the solution to the system $2x + y = 4$ and $x - 2y = -3$, when solved by graphing?
(A) $(2, 0)$
(B) $(0, 4)$
(C) $(-2, 0)$
(D) $(1, 1)$
(E) $(0, -2)$

Q7. In a study on the effect of CO_2 levels on plant growth, the rate of growth r is modeled by the equation $r = -2x + 10$, where x is the CO_2 level. What is the x-intercept of this line?
(A) $(0, 10)$
(B) $(5, 0)$
(C) $(3, 0)$
(D) $(0, -2)$
(E) $(10, 0)$

Q8. A patient's dosage D of a certain medication is given by $D = 0.5x + 2$, where x is the patient's weight. What is the y-intercept of this line?
(A) $(0, 2)$
(B) $(0, 0.5)$
(C) $(2, 0)$
(D) $(0, -2)$
(E) $(2, 2)$

Open Ended Questions (FRQ)

Q9. Graph the system $y = x - 2$ and $y = -x + 4$. Find the intersection point and interpret its meaning in the context of a medical trial where x represents the dosage of a medication and y represents the patient's response.

Q10. The population growth of two types of cells can be modeled by the system: $y = 2x + 1$ and $y = x - 3$. Solve this system by graphing and interpret the results in the context of cell biology, where x represents time and y represents the cell population.

Chef's Tips

- Tip 1: Ensure students use graph paper to plot the lines accurately.
- Tip 2: Have students check their intersection points algebraically to verify their graphical solutions.
- Tip 3: Emphasize the importance of interpreting the intersection point in the context of the given problem.